
Algorithm 1: EGL — coverage-guided line localization (one instance). Execution coverage E is a *membership* signal injected at three stages of a vote-based edit-location step running on a pluggable localization host: **(1)** upstream pool expansion (raise the superset ceiling), **(2)** midstream conditioned generation, **(3)** downstream frontier filter (cut LoC). The output is an *unordered* candidate set, scored by SUPERSET (predicted set $\supseteq$ all gold lines) and LoC (set size). $E^{\pm\delta}$ is E dilated by a $\pm\delta$ line window; E_f the executed lines of file f .

Require: issue I ; repo $\mathcal{R}$; failing-test coverage E ; host localizer; samples k ; expansion budget K ; frontier width δ ; cut-off N ; window w

Result : candidate set $\mathcal{S}$; report $\text{SUPERSET} = [\text{gold} \subseteq \mathcal{S}]$ and $\text{LoC} = |\mathcal{S}|$

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1 ( $\mathcal{F}, \text{Rel}$ )  $\leftarrow$  HOSTLOCALIZE( $I, \mathcal{R}$ )            $\triangleright$  pluggable host: candidate files + related elements
2  $\text{voted} \leftarrow \{\}$                                 $\triangleright \{(\text{file}, \text{line}) : \text{freq}\}$ 
3 foreach file  $f$  with elements  $\text{Rel}_f$  do
4    $\Omega \leftarrow \text{LINES}(\text{Rel}_f, \pm w)$             $\triangleright$  coarse element region
5    $\Omega \leftarrow \Omega \cup \text{SCOREDEXECTOPK}(f, I, E_f, K)$     $\triangleright$  (1) top- $K$  executed lines by LINESCORE
      (issue + traceback slice + coverage); ceiling 68.4  $\rightarrow$  75.4
6    $\text{freq} \leftarrow \mathbf{0}$ 
7   for  $s \leftarrow 1$  to  $k$  do
8      $L_s \leftarrow \text{LLM}(I, \text{VIEW}(\Omega, E_f), \tau_s)$     $\triangleright$  (2) conditioned generation: mark executed ‘>’ +
      prepend crash traceback
9     foreach  $(r, \ell) \in L_s$  do  $\text{freq}[\ell] += 1/(1 + r)$ 
10  end
11  foreach  $\ell$  do  $\text{voted}[(f, \ell)] \leftarrow \text{freq}[\ell]$ 
12 end
13  $\text{voted} \leftarrow \{(f, \ell) \in \text{voted} : \ell \in E_f^{\pm\delta}\}$     $\triangleright$  (3) downstream frontier filter  $\rightarrow$  cut LoC
14  $\mathcal{P} \leftarrow \text{TOPN}(\text{voted}, N)$             $\triangleright N$  highest-frequency pairs (unordered)
15  $\mathcal{S} \leftarrow \bigcup_{(f, \ell) \in \mathcal{P}} \text{WINDOW}(f, \ell, \pm w)$     $\triangleright \pm w$  expansion; the reported edit-location set
16 return  $\mathcal{S}$             $\triangleright \text{SUPERSET} = [\text{gold} \subseteq \mathcal{S}], \text{LoC} = |\mathcal{S}|$ ; sweep  $N$  for the Pareto curve

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